

Algebra II with Trigonometry

Chapter 5.6

Olympiad Problems (GPT-5.4)

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[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

Problems

1. A triangle has sides 13, 14, 15. Find the largest angle.
2. Two sides of a triangle are 20 and 21, and the included angle is 60° . Find the third side, then the area.
3. A triangle has sides 25, 25, 30. Find its area without using a calculator.
4. A hiker walks 12 km due east, then $5\sqrt{2}$ km on a bearing $N45^\circ E$. How far is the hiker from the starting point?
5. Point B is 100 miles from point A at bearing $N60^\circ E$. A pilot mistakenly flies due east from A for 50 miles. How far is the pilot from B ?
6. A triangle has sides 29, 29, 42. Find the exact value of $\cos \theta$, where θ is one of the base angles.